Odd Cycles with One Pendant Leaf: Fractional Degree Bias and Hölder Compression

Abstract

Let $m \geq 3$ be odd and let C_m^+ be obtained from the odd cycle C_m by attaching one pendant leaf at one cycle vertex. We prove that every graphon W of edge density $p \geq 1/2$ satisfies

$$t(C_m^+, W) \geq p\big(p^m - p(1-p)^{m-1}\big) = \Phi_{C_m^+}(p).$$

The proof averages the leaf root over the cycle, applies the arithmetic– geometric mean inequality, and changes the underlying probability measure by the fractional degree weight $d^{1/m}$. A sharp three-factor Hölder inequality controls the new edge density, while a one-variable monotonicity lemma transfers the accepted odd-cycle bound without loss. The case $m = 5$ classifies the formerly open NetworkX Atlas graph 104.

1 Definitions and the exact target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y).$$

For odd $m \geq 3$, label the vertices of C_m cyclically by $0, \ldots, m-1$. Let C_m^+ be the graph obtained by adding one new leaf adjacent to vertex 0. The chromatic polynomial of an odd cycle is

$$\chi_{C_m}(x) = (x-1)^m - (x-1).$$

The leaf has $x-1$ choices after its neighbour is coloured, so

$$\chi_{C_m^+}(x) = (x-1)^{m+1} - (x-1)^2, \qquad \chi(C_m^+) = 3. \tag{1}$$

Consequently, with $q = 1-p$,

$$\begin{aligned} \Phi_{C_m^+}(p) &= p\big(p^m - pq^{m-1}\big) \\ &= p^{m+1} - p^2(1-p)^{m-1}. \end{aligned} \tag{2}$$

This formula is polynomial and gives the continued value one at $p = 1$.

We use the accepted odd-cycle graphon theorem in the following exact form: for every odd $m \geq 3$, every graphon K on every probability space, and its edge density s,

$$t(C_m, K) \geq \varphi_m(s), \qquad \varphi_m(s) := s^m - s(1-s)^{m-1}. \tag{3}$$

This is Theorem 1.4 of the paper (Im, Kim, and Yang). We do not assume any rooted strengthening of it.

2 Fractional edge Hölder inequality

The next lemma is useful independently of cycles.

Lemma 2.1 (Fractionally degree-weighted edge)**.** *Let $0 < \alpha \le 1$ and put*

$$N_\alpha = \int_{\Omega^2} W(x,y) d(x)^\alpha d(y)^\alpha \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y).$$

Then

$$N_\alpha \ge p^{1+2\alpha}.$$

In particular, for $\alpha = 1/m$,

$$N_{1/m} \ge p^{1+2/m}.$$

Proof. On the set where $d(x) = 0$, one has $W(x,y) = 0$ for almost every y. Define $W(x,y)/d(x)$ to be zero there. Put

$$\theta = \frac{1}{1+2\alpha}, \qquad \eta = \frac{\alpha}{1+2\alpha}.$$

Then $\theta + 2\eta = 1$, and almost everywhere on the support of W,

$$W(x,y) = \big(W(x,y) d(x)^\alpha d(y)^\alpha\big)^\theta \left(\frac{W(x,y)}{d(x)}\right)^\eta \left(\frac{W(x,y)}{d(y)}\right)^\eta.$$

Weighted Hölder gives

$$p \le N_\alpha^\theta \left(\int \frac{W(x,y)}{d(x)} \, \mathrm{d}\mu^2\right)^\eta \left(\int \frac{W(x,y)}{d(y)} \, \mathrm{d}\mu^2\right)^\eta.$$

Each quotient integral is at most one: integrating first in the other variable gives one on $\{d > 0\}$ and zero on $\{d = 0\}$. Hence $p \le N_\alpha^\theta$, which is the claimed inequality. □

3 The scalar transfer

Lemma 3.1 (Odd-cycle rescaling)**.** *Let $m \ge 3$, $p \in [1/2, 1]$, and $0 < z \le p$. Set*

$$s_0 = p \left(\frac{p}{z}\right)^{2/m}.$$

If $s_0 \le 1$, then

$$z \, \varphi_m(s_0) \ge p \, \varphi_m(p).$$

Proof. For $s \in (0,1]$, put

$$\psi_m(s) = \frac{\varphi_m(s)}{s^{m/2}} = s^{m/2} - s^{1-m/2}(1-s)^{m-1}.$$

Differentiation gives

$$\psi_m'(s) = \frac{m}{2} s^{m/2-1} + (m-1) s^{1-m/2}(1-s)^{m-2} + \left(\frac{m}{2} - 1\right) s^{-m/2}(1-s)^{m-1} > 0.$$

Thus ψ_m is strictly increasing. Write $r = z/p \le 1$. Then $s_0 = p r^{-2/m} \ge p$ and $r = (p/s_0)^{m/2}$. Therefore

$$z\varphi_m(s_0) = p^{1+m/2} \psi_m(s_0) \ge p^{1+m/2} \psi_m(p) = p\varphi_m(p).$$

□

We shall also use that φ_m is increasing on $[1/2, 1]$. Indeed,

$$\varphi_m'(s) = ms^{m-1} - (1-s)^{m-1} + (m-1)s(1-s)^{m-2} > 0$$

there, since $ms^{m-1} \geq m(1-s)^{m-1}$.

4 The one-leaf odd-cycle theorem

Theorem 4.1 (One leaf on an odd cycle)**.** *For every odd integer $m \geq 3$ and every graphon W of edge density $p \in [1/2, 1]$,*

$$\boxed{t(C_m^+, W) \geq p\big(p^m - p(1-p)^{m-1}\big) = \Phi_{C_m^+}(p).}$$

Proof. Let

$$F(x_0, \ldots, x_{m-1}) = \prod_{i=0}^{m-1} W(x_i, x_{i+1}), \qquad x_m = x_0.$$

Integrating the pendant leaf first gives

$$t(C_m^+, W) = \int F\, d(x_0)\, \mathrm{d}\mu^m.$$

Cyclic relabelling shows that the same density results when the degree factor is placed at any x_i. Averaging these m equal integrals and applying arithmetic–geometric mean pointwise gives

$$t(C_m^+, W) \geq \int F \prod_{i=0}^{m-1} d(x_i)^{1/m}\, \mathrm{d}\mu^m. \tag{4}$$

Put

$$M = \int_\Omega d(x)^{1/m}\, \mathrm{d}\mu(x).$$

Since $p \geq 1/2$, one has $M > 0$. Define the probability measure

$$\mathrm{d}\nu(x) = \frac{d(x)^{1/m}}{M}\, \mathrm{d}\mu(x).$$

The right side of (4) is

$$M^m t(C_m, W; \nu), \tag{5}$$

where the notation means that all vertex variables are integrated using ν. The edge density of W on this new probability space is

$$s = \frac{N_{1/m}}{M^2}.$$

Set $z = M^m$. Concavity of $r \mapsto r^{1/m}$ gives

$$M \leq p^{1/m}, \qquad 0 < z \leq p.$$

Lemma 2.1 gives

$$s \geq \frac{p^{1+2/m}}{M^2} = p\left(\frac{p}{z}\right)^{2/m} =: s_0.$$

Since $W \le 1$ and ν is a probability measure, $s \le 1$; hence also $s_0 \le 1$. In particular $s \ge s_0 \ge p \ge 1/2$.

Apply the accepted odd-cycle theorem (3) on (Ω, ν), monotonicity of φ_m, and Lemma 3.1:

$$\begin{aligned} t(C_m^+, W) &\ge M^m t(C_m, W; \nu) \\ &\ge z\varphi_m(s) \ge z\varphi_m(s_0) \ge p\varphi_m(p). \end{aligned}$$

The last expression is exactly (2). □

At $p = 1/2$ the target is zero, while every step remains valid. At $p = 1$, $W = 1$ almost everywhere and both sides equal one. No division by $2p - 1$ or by a cycle density occurs.

Every balanced complete k-partite graphon, $k \ge 2$, is regular of degree $p = 1 - 1/k$. Then $M = p^{1/m}$, $z = p$, $s = p$, and equality holds in every step precisely when it holds in the accepted odd-cycle theorem. Thus the new theorem has the expected chromatic equality examples.

5 Catalogue consequence and formalization boundary

Corollary 5.1 (Atlas 104)**.** *NetworkX Atlas graph* 104*, with graph6 code* `Ehd?`*, is positive. It has order six, size six, chromatic number three, and*

$$\chi_{F_{104}}(x) = (x-1)^6 - (x-1)^2.$$

Every graphon of edge density $p \in [1/2, 1]$ satisfies

$$t(F_{104}, W) \ge p^6 - p^2(1-p)^4 = p^2(2p-1)(2p^2 - 2p + 1).$$

Proof. Its NetworkX edge list is

$$01,\ 04,\ 12,\ 15,\ 23,\ 34.$$

Vertices $0, 1, 2, 3, 4$ induce the cycle $0 - 1 - 2 - 3 - 4 - 0$, and vertex 5 is a single leaf adjacent to vertex 1. Thus it is C_5^+, and Theorem 4.1 applies. □

The proof uses ordinary, non-injective homomorphism densities. All integrands are bounded and measurable, so Tonelli–Fubini applies on an arbitrary probability space. The quotient W/d is explicitly defined to be zero on $\{d = 0\}$, where W vanishes in the relevant sections almost everywhere. No finite-step reduction, injective-copy interpretation, regularity, or minimum-degree assumption is used.

For formal verification, the reusable new input is Lemma 2.1, followed by the measure change (5) and the elementary scalar Lemma 3.1. The existing odd-cycle theorem is used with the same graphon but the biased probability measure ν. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph104.lean`.

The method covers exactly one pendant leaf on an odd cycle. With two or more leaves, averaging gives a different fractional degree product and the present scalar transfer does not apply automatically. It also does not prove leaf attachment to an arbitrary positive graph: cyclic symmetry is used essentially in (4).

The chromatic and scalar identities, the fractional Hölder exponents, and the Atlas identification were audited in exact rational arithmetic.

6 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/OddLeaf/`, ending at `satisfiesLowerBound_104`.

Scope: $m = 5$ only. Theorem 4.1 is *not* formalized for general odd m. What is proved is Corollary 5.1, on all of $p \in [1/2, 1]$, and it is the only member of the family inside the six-vertex catalogue. The reason is (3): the project's checked odd-cycle input is the five-cycle theorem `OddCycleC5.c5_homDensity_bound`, not a general odd m. The degree-bias construction is nonetheless parametric — the file `OddLeaf/Bias.lean` builds $d^{1/m}$, $M = \int d^{1/m}$, the tilted measure ν and the reread kernel for every $m \geq 1$ — while `OddLeaf/Holder.lean` and the scalar rescaling are specialized to $m = 5$.

The departure: the rescaling carries no derivative. Lemma 3.1 is proved here by differentiating $\psi_m(s) = \varphi_m(s)s^{-m/2}$, a function with two half-integer powers. The formalization forms no half-integer power and differentiates nothing. Write

$$\varphi_5(u) = u^5 - u(1-u)^4 = u\,h(u), \qquad h(u) = (2u-1)(2u^2-2u+1),$$

which is `OddLeaf.phi_eq`, a `ring` identity. All of Lemma 3.1 then reduces to the polynomial statement

$$s^3 h(p)^2 \leq p^3 h(s)^2 \qquad \big(\tfrac{1}{2} \leq p \leq s\big), \tag{6}$$

which is `OddLeaf.scalar_mono`. Substituting $p = \frac{1}{2} + a^2$ and $s = \frac{1}{2} + a^2 + b^2$ — legitimate because the two hypotheses are exactly $p - \frac{1}{2} \geq 0$ and $s - p \geq 0$, so both differences are squares of their own square roots — the cleared form $8\big(p^3 h(s)^2 - s^3 h(p)^2\big)$ becomes a polynomial in a, b with 36 terms, every coefficient positive and every exponent even. One `ring` identity and `positivity` finish it.

Monotonicity of φ_m is never used. The unnumbered remark after Lemma 3.1 — that $\varphi_m' > 0$ on $[1/2, 1]$, used in the proof of Theorem 4.1 to pass from s to s_0 — is not formalized. It is not needed. Writing $K = M^5$ and letting s be the biased edge density, `OddLeaf.transfer` proves

$$p\,\varphi_5(p) \leq K\,\varphi_5(s)$$

directly from the three facts $0 < K \leq p$, $p^7 \leq K^2 s^5$ and (6), by comparing squares:

$$\big(K\varphi_5(s)\big)^2 s^5 p^5 = \big(K^2 s^5\big)\big(p^5 \varphi_5(s)^2\big) \geq p^7\big(s^5 \varphi_5(p)^2\big) = \big(p\varphi_5(p)\big)^2 s^5 p^5.$$

The intermediate quantity $s_0 = p(p/z)^{2/m}$ therefore never appears, and neither does the hypothesis $s_0 \leq 1$. The inequality $p \leq s$, which the note gets from $z \leq p$ and the definition of s_0, is obtained instead from $p^7 \leq K^2 s^5 \leq p^2 s^5$.

Two statements are kept polynomial on purpose. Lemma 2.1 is formalized as

$$p^7 \leq N^5 \qquad (\texttt{OddLeaf.pow_seven_le_pow_rootEdge}),$$

not as $N \geq p^{7/5}$, so that no real power of p is ever formed; the $\mathbb{R}_{\geq 0}^{\infty}$ Hölder step itself is exactly the note's, at exponents $(5/7, 1/7, 1/7)$, and it reuses the scaffolding of `Methods/PathSidorenko.lean` — the pushforwards `edgeE`, `degE`, the two bounds $\iint W/d \leq 1$, and the null set where the factorization breaks — verbatim. Likewise the concavity bound $M \leq p^{1/m}$ of the main proof is formalized as $M^m \leq p$ (`OddLeaf.pow_rootMean_le`), by Jensen against the *convex* $t \mapsto t^m$ rather than by concavity of $r \mapsto r^{1/m}$. This is why the exponent is fixed to a reciprocal integer: it keeps both $(d^{1/m})^m = d$ and $M^m \leq p$ as ordinary $\mathbb{N}$-powers.

The cyclic averaging is a measure-preserving permutation. "Cyclic relabelling shows that the same density results when the degree factor is placed at any x_i" is discharged by `Methods/Peeling.integral_assignment_perm`, a new general lemma extracting the measure-preserving half of `homDensity_iso` so that it applies to an integrand which is *not* a graph weight — here the cycle weight multiplied by a degree at one coordinate. It is applied to the rotation $j \mapsto j + i$, checked to be an automorphism of C_5 by `decide`. The arithmetic–geometric mean step (4) is Mathlib's `Real.geom_mean_le_arith_mean_weighted` at the five equal weights $1/5$, and the measure change (5) goes through `Foundation/TiltTransfer.lean`, the same route as the d- and $\sqrt{d}$-biases already in the project.